

Siddhartha Phuyal

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EDUCATION

- **University of South Dakota** Vermillion, SD
Master of Science in Computer Science; GPA: 3.88 *Jan. 2024 – Aug. 2025*
- **The British College** Kathmandu, Nepal
Master of Business Administration *Sept. 2022 – Aug. 2024*
- **KIIT University** Bhubaneswar, India
Bachelor of Technology in Civil Engineering; GPA: 7.44/10.0 *Aug. 2016 – Jul. 2020*

WORK EXPERIENCE

- **University of South Dakota** Vermillion, SD
Student *Jan. 2024 – Present*
 - **TensorFlow:** Worked on deep learning model development using TensorFlow, an open-source software library for numerical computation.
 - **Keras:** Built and trained neural networks using Keras, a high-level API that integrates with TensorFlow.
- **Golden Goods Construction Private Ltd.** Dorpa, Khotang, Nepal
Site Engineer *Dec. 2020 – May 2022*
 - **Project Management:** Directed engineering activities at Mid-hill Highway project sites, including scheduling daily work for a 20+ member crew, coordinating with contractors, and monitoring progress to ensure timely completion of 10 km of roadway.
 - **Engineering Solutions:** Designed and supervised slope stabilization measures using gabion retaining walls and improved drainage systems to address frequent landslides and water-logging issues along the highway.
 - **Testing and Observation:** Conducted soil classification and compaction tests, observed concrete curing, and verified asphalt layer thickness to ensure compliance with Nepal Road Standards.
- **Green Apple Incorporation** Kathmandu, Nepal
Educational Consultant *Dec. 2022 – Jul. 2023*
 - **Student Counseling:** Counseled students on overseas education programs and assisted with visa applications.

PROJECTS

- **Binary Image Classification:** Built a binary image classification model using TensorFlow's Keras API, tailored for medical imaging (Chest X-rays).
- **Feature Extraction and Fine-Tuning:** Performed feature extraction and fine-tuning using VGG19 and ResNet50 on skin cancer image data.
- **Diabetes Risk Prediction:** Analyzed and evaluated machine learning models (Decision Trees, Random Forest, XGBoost, LightGBM, CatBoost) for predicting diabetes risk.
- **Multimodal Model Fusion:** Designed a multimodal model fusion framework combining MRI images and textual patient data for breast lesion analysis.
- **CNN:** Developed a text mining solution for cancer care using clinical medical records, leveraging Convolutional Neural Networks (CNN).
- **Vehicle Number Plate Detection:** Implemented vehicle number plate detection using YOLO v8 for object detection and OCR (Optical Character Recognition) for state identification.

PROGRAMMING SKILLS

- **Languages:** Python, JavaScript, C

CERTIFICATIONS

- **Supervised Machine Learning: Regression and Classification:** An online non-credit course authorized by DeepLearning.AI and Stanford University, offered through Coursera.
- **Advanced Learning Algorithms:** An online non-credit course authorized by DeepLearning.AI and Stanford University, offered through Coursera.
- **Macroeconomics:** Udemy certificate in "The Macroeconomics."
- **Financial Modeling and Valuation in Microsoft Excel:** Udemy certificate in "Financial Modeling and Valuation in Microsoft Excel."
- **Civil Engineering License:** Licensed Civil Engineer from the Nepal Engineering Council.
- **Vedic Astrology:** Vedic Astrology certificate from Magha Vedic Astrology Academy.

COURSES STUDIED IN GRADUATE LEVEL

- **Machine Learning Fundamentals:** Studied core concepts in supervised and unsupervised learning, model evaluation, and optimization techniques.
- **Computer Vision:** Focused on image processing, object recognition, and deep learning applications in visual data.
- **Data Mining:** Explored methods for discovering patterns and extracting insights from large datasets.
- **Computer Networks and Security:** Learned about networking protocols, data transmission, and cybersecurity practices.
- **Design and Analysis of Algorithms:** Studied algorithm design, complexity analysis, and optimization techniques.
- **Applied Reinforcement Learning:** Worked on practical applications of reinforcement learning for solving real-world problems.
- **Applied Math for Data Science and Machine Learning:** Applied advanced mathematical concepts, such as linear algebra and calculus, in data science and machine learning contexts.
- **Corporate Finance:** Studied financial decision-making processes, including valuation, risk management, and capital structure.
- **Marketing Dynamics:** Focused on customer behavior, market segmentation, and strategic marketing techniques.
- **Operations and Logistics Management:** Studied supply chain management, inventory control, and optimization of operational efficiency.

LANGUAGES

- **Nepali:** Native proficiency.
- **English:** IELTS: Reading 7.5, Writing 7, Speaking 7, Listening 6, Overall 7.
- **Hindi:** Fluent in both understanding and speaking.

REFERENCES

- **Assistant Prof. Dr. Debesh Jha:** Visiting Assistant Professor, University of South Dakota
Phone: 605-658-6840
Email: debesh.jha@usd.edu
Department of Computer Science
- **Prof. Dr. KC Santosh:** Chair and Professor, Department of Computer Science, University of South Dakota
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- **Assistant Prof. Dr. Rodrigue Rizk:** Assistant Professor, University of South Dakota
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